

bodySITARA

Privacy-Preserving Video Anonymization Pipeline

A Complete, Plain-English Guide to the ComfyUI Workflow

Workflow file: V4.4 — b1 active, no blockify

What this document explains:

the complete flow in everyday language • every node's inputs and outputs
where each result goes next • every adjustable setting
and its effect on quality and on speed

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Part 1 — The Whole Idea in Plain Words

Before any technical detail, here is the entire journey told as a simple story. No node names, no jargon — just what happens to your video, step by step. Read this once and the rest of the document will make sense.

The story of one video clip

- 1. A video comes in.** You give the system a short clip of a person moving around (walking, gesturing, talking).
- 2. The system finds the person.** It looks at every frame and draws an invisible outline (a “cut-out shape”) tightly around the person — like tracing them with a marker. This outline is called a mask.
- 3. The outline is made a little bigger and its edges softened.** The cut-out shape is grown slightly so it safely covers the whole body (hair, fingertips, loose clothing) and its border is blurred so later steps blend nicely instead of showing a hard cookie-cutter line.
- 4. The person is hidden.** Everywhere inside that outline, the real person is painted over with a flat grey shape. The background stays untouched. Now you have a “privacy video”: the scene is normal, but the human is just a moving grey silhouette. Nobody can tell who they are.
- 5. Two harmless ‘notes’ about the person are written down.** At the same time, the system records two things that describe how the person moves without revealing who they are: (a) a stick-figure skeleton showing their pose in each frame, and (b) a simplified face pattern that captures their expression (smiling, blinking, head turning) but not their actual face. Think of these as choreography notes and acting notes.
- 6. The person is sliced out from the background.** The grey silhouette is separated from the scene. Now there are two separate things: the grey person-shape on its own, and the background with a person-shaped hole in it.
- 7. The hole in the background is filled in.** The system looks at other moments in the video — frames where the person had moved aside — to figure out what the wall, floor, or furniture behind them actually looks like, and paints that back in. The result is a clean, empty background as if the person were never there. Any small leftover gaps are patched by a smart ‘fill-in’ tool.
- 8. A brand-new stand-in actor is generated.** This is the clever part. A powerful AI video generator is given: a single reference photo of a stand-in person, the skeleton choreography notes, and the face-expression acting notes. From these it performs the same movements and expressions — but as a completely different, made-up human. The original person’s identity is gone; only their motion and expression survive, now worn by someone else.
- 9. The stand-in actor is placed onto the clean background.** The newly generated person is cut out and laid on top of the filled-in background, lined up exactly where the original person was. Their edges are softened so they sit naturally in the scene.
- 10. The finished video is saved (with the original sound).** You now have a final clip: the same scene, the same movements and expressions, the same audio — but performed by a

synthetic person who never existed. Privacy is protected, yet the video still looks and behaves like real footage.

11. A comparison strip is made (for testing). Finally, for evaluation, the system can place all the versions side by side — original, grey-silhouette, empty background, generated actor, and final — so you can judge how well each stage worked.

The one-sentence summary: *take a real person out of a video, remember only how they moved and emoted, rebuild the empty scene, and drop in a fake person who copies the motion — so the footage stays useful but the real human can never be identified.*

Part 2 — Why It Is Built This Way (the privacy logic)

This workflow is not just “blur a face.” It is a privacy pipeline designed so that the secret, identifying information never leaves a place you trust. To understand the node layout, you only need three ideas.

Idea 1: Some computers are trusted, some are not

The pipeline imagines the work being split across different machines with different levels of trust. Anything that can reveal who the person is must be handled on a trusted device (in the project’s framing, the smart-glasses and the user’s own phone). The heavy AI generation runs on an untrusted cloud machine — powerful, but not allowed to ever see the real face. That is why the work is divided into labelled “Tiers,” and why you will see notes like “PORT OUT” and “PORT IN” marking exactly what data is allowed to cross from a trusted device to the cloud.

Idea 2: Only motion and expression cross the boundary — never identity

When the package is sent to the cloud, it contains just three things: the grey silhouette video (no face), the stick-figure pose, and the abstract face-expression pattern. The cloud receives enough to animate a body realistically, but nothing it could use to recover the original person’s appearance. The real face and the original pixels stay behind on the trusted side.

Idea 3: Rebuild, don’t reveal

Instead of trying to ‘un-blur’ or lightly disguise the real person (which can often be reversed), this design throws the real person away entirely and rebuilds a new one from scratch. The background is reconstructed; the human is regenerated. There is no hidden original underneath to leak. This is what makes the approach strong: privacy comes from replacement, not from hiding.

The six labelled regions (Tiers) on the canvas

Region (group label)	Where it runs	Its single job
TIER 1 — Glasses (trusted)	Trusted capture device	Find the person, hide them under grey, and write the pose + face-expression notes.
TIER 2 Part A — Mobile	Trusted phone	Separate the grey person from the scene and package it for the cloud.
TIER 2 Part B-1 — Mobile	Trusted phone	Erase the person from the scene and fill the hole to make a clean empty background.
TIER 2 Part B-2 — Cloud (untrusted)	Powerful cloud GPU	Generate a brand-new synthetic person performing the recorded motion + expression.
TIER 2 Part C — Mobile	Trusted phone	Composite the generated person onto the clean background = the final video.

Region (group label)	Where it runs	Its single job
Evaluation only	Anywhere (testing)	Lay all stages side by side to judge quality. Not part of the real privacy product.

Part 3 — How to Read the Technical Sections

The rest of this document walks through the actual building blocks (“nodes”) in order. A few quick concepts will let you read every page that follows.

Nodes and wires

A node is a single box that does one job. A wire carries the output of one node into the input of another. Data only flows in the direction of the wire. So the whole workflow is just boxes connected by wires — like an assembly line where each station does one task and passes the result down the belt.

Inputs vs. settings (the two ways to feed a node)

- **An input** is fed by a wire from another node (e.g. “the video frames”).
- **A setting (also called a widget or parameter)** is a value you type into the node itself (e.g. “blur radius = 18”). These settings are exactly the ‘knobs you can turn’, and this document lists them for every node along with what they do to quality and speed.

Important rule: if a setting also has a wire plugged into it, the wire wins and the typed value is ignored. You will see this happen with width/height below — the nodes show ‘832 × 480’ typed in, but a wire overrides them to the real working size of 640 × 640.

‘Set’ and ‘Get’ boxes = invisible wires

You will see many small boxes named Set_something and Get_something. These are not real processing steps — they are a tidiness trick. A ‘Set’ box saves a result under a name (like saving a file), and a matching ‘Get’ box anywhere else on the canvas hands that same result back out. This avoids drawing long wires across a giant graph. Whenever you see Get_filled_background, just read it as “the filled background that was produced earlier.” The boxes labelled PORT OUT and PORT IN are these same Set/Get pairs, used here to mark the exact data that crosses from a trusted device to the cloud.

What a ‘mask’ is

A mask is a black-and-white stencil the same size as the video. White means “this area is selected” and black means “leave this alone.” Almost every privacy step in this workflow is really just: make a stencil of the person, then do something only inside (or only outside) that stencil — hide them, cut them out, or punch a hole. If you understand stencils, you understand most of this pipeline.

A note on this particular file

The file is named “V4.4 — b1 active, no blockify.” Two things this tells you, both confirmed inside the graph:

- **“b1 active”** means the background-filling branch (Tier 2 Part B-1) is switched on and feeding the final result.

- **“no blockify”** means the pixelation step in Tier 1 is switched off. (A separate pixelation step inside the cloud branch is still on — more on this when we reach it.)

Several nodes in the file are deliberately switched off (bypassed). They are older or alternative versions of a step, kept on the canvas as spare parts. They are all listed together near the end so you know what is genuinely running versus what is dormant.

Part 4 — Stage-by-Stage Walkthrough

Each node below is presented the same way: a plain-words purpose, what it takes in (and from where), what it sends out (and to where), and a table of every knob you can turn with its effect on quality and on speed. Node ID numbers (e.g. [27]) match the numbers in the actual file so you can find them on the canvas.

Stage 0 — Shared Setup: loaders and the master resolution

These nodes do not process the video by themselves; they load the inputs and the AI models that the rest of the pipeline relies on, and they set the resolution everything runs at.

[150] / [151] Width / Height — the master resolution dials

***In plain words:** Two little number boxes that decide the working resolution for the whole pipeline. They are wired into the video loader, the reference-image resizer, and the detection nodes, so changing these two numbers changes the size everything runs at — a single, central control point.*

Sends out →

- The value 640 (Width) and 640 (Height) into the video loader, the resize node, and pose detection.

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
value (currently 640)	The pixel size every stage works at. This is THE main lever for the whole run. It overrides the 832×480 typed into other nodes.	Higher = sharper, more facial/clothing detail in the generated person. Lower = softer, blockier.	Cost grows roughly with the square of the size: doubling it makes generation ~4× slower and uses far more GPU memory. Lower it first when testing.

[63] Load Video (VHS_LoadVideo)

***In plain words:** Reads your input clip from disk and turns it into a sequence of image frames the pipeline can work on. Also extracts the audio so the final video can keep its sound.*

Takes in ←

- **custom width / height** from the master dials [150]/[151].

Sends out →

- **the frames** (saved as 'input_video', used by Tier 1).
- **the frame count** (saved as 'frame_count'; tells later nodes how many frames to make).
- **the audio** (saved as 'input_audio'; re-attached at the very end).

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
force_rate (10)	Re-times the clip to this many frames per second before anything else. The	Higher fps = smoother motion. Lower fps = choppy	Fewer fps = fewer frames = the whole pipeline runs

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
	single biggest control over how many frames exist.	but identical per-frame quality.	proportionally faster and lighter. 10 is a deliberately low, fast setting.
frame_load_cap (0)	Maximum number of frames to load. 0 means 'all of them'.	No quality effect; just length.	Set this to a small number (e.g. 16) for quick test runs, then back to 0 for the full clip. Huge time saver while experimenting.
skip_first_frames (0) / select_every_nth (1)	Skip frames at the start, or take only every Nth frame.	Subsampling makes motion choppy.	Both reduce frame count = faster.
custom_width / height (overridden)	Resolution to load at — but a wire from the master dials overrides these to 640×640.	See master dials.	See master dials.

[57] Load Image (LoadImage)

***In plain words:** Loads the single reference photo of the stand-in person — the face/body the AI will use to build the synthetic actor. Choosing a clear, well-lit, front-facing photo here strongly affects how good the generated person looks.*

Sends out →

- the reference photo to the resizer [64].

No quality/speed knobs — but the choice of photo itself is one of the most important quality factors in the whole workflow.

[64] Resize Reference Image (ImageResizeKJv2)

***In plain words:** Resizes the reference photo to the pipeline's working resolution so it lines up with everything else.*

Takes in ←

- **the reference photo** from [57]; **width/height** from the master dials.

Sends out →

- the resized reference (saved as 'reference_image'), used by the cloud generator and its image encoder.

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
upscale_method (lanczos)	The maths used when resizing. Lanczos keeps edges crisp.	Lanczos = sharper than simpler methods. Minor effect.	Negligible.

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
keep_proportion (pad_edge_pixel)	How to fit the photo into the target shape. 'Pad' adds border pixels instead of stretching, so the face is not distorted.	Prevents the squashed/stretched look. Good for faces.	None.
divisible_by (16)	Rounds the size to a multiple of 16 because the AI model requires it.	Avoids errors / artefacts at edges.	None.

The AI model loaders (grouped)

Several boxes simply load large AI models into memory once, then hand them to the working nodes. They are described briefly together because you rarely change them frame-to-frame; the values shown are sensible defaults for the project's GPU.

Loader node	What it loads / does	Knobs that matter for quality ↔ speed
[22] WanVideoModelLoader	Loads the main Wan 2.2 Animate video-generation model — the 'brain' that creates the synthetic person.	base_precision (fp16_fast): faster, tiny quality cost. quantization (disabled): turning it on (e.g. fp8) saves GPU memory at a slight quality cost. attention_mode (sdpa): the algorithm for the model's attention; alternatives can be faster if installed.
[38] WanVideoVAELoader	Loads the 'translator' (VAE) that converts between the model's compressed internal sketch and real pixels.	precision (bf16): standard. Rarely changed.
[71] CLIPVisionLoader + [70] WanVideoClipVisionEncode	Reads the reference photo into a form the generator understands as 'who to look like'.	strength (0.5): how strongly the reference appearance guides the result. Higher = sticks closer to the photo; too high can look pasted-on.
[178] OnnxDetectionModelLoader	Loads the person-detector and full-body pose model used to find the person and read their skeleton.	Bigger models (vitpose-l, yolov10m) = more accurate detection, a little slower. Smaller models = faster, less precise.
[102] DownloadAndLoadSAM2Model	Loads SAM 2, the model that draws the precise person-shaped stencil (mask).	base_plus model = good accuracy/speed balance; larger variants are more precise but slower.
[264] INPAINT_LoadInpaintModel	Loads 'big-lama', a smart patch-filler used to repair leftover holes in the background.	Fixed model; no real quality/speed dial.
[65] WanVideoTextEncodeCached	Turns the text prompt into guidance for the generator. Positive prompt is empty (the reference photo does the describing); the negative prompt lists things to avoid (over-saturation, blur, subtitles, worst-quality, etc.).	Editing the positive prompt can nudge style/quality; the negative prompt suppresses common defects.

Tier 1 — Glasses (trusted): find, hide, and describe the person

This is the most important privacy stage. It runs on the trusted device. By the end of it, the real person is gone (replaced by grey), and two identity-free ‘notes’ about their motion and expression have been written. Everything downstream works only from these safe outputs.

[273] Pose + Face Detector (RTMPoseTinyPoseAndFace)

In plain words: Looks at the frames and finds three things at once: a box around the person, the full-body skeleton key-points (shoulders, elbows, knees, etc.), and a tight crop of the face. It is the eyes of Tier 1 — everything else builds on what it finds.

Takes in ←

- **the video frames + their size** from [180] (which reads the saved input video).

Sends out →

- **skeleton key-point data** to the pose drawer [173].
- **the face crop** to the face-mesh maker [192].
- **the person’s bounding box** to the stencil maker SAM 2 [104].

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
width / height (832×480)	Resolution the detector analyses at.	Higher = catches finer/ smaller key-points. Lower = may miss fingers/face detail.	Higher = slower detection.
confidence thresholds (0.3 / 0.3 / 0.4)	How sure the model must be before it accepts a body point / hand point / face point.	Higher = only confident points kept (cleaner but may drop some). Lower = more points but noisier, can jitter.	Negligible.

[104] Make the Person Stencil (Sam2Segmentation)

In plain words: Takes the rough box around the person and turns it into a precise, pixel-perfect cut-out shape of the body (the mask/stencil). This is the shape that will be hidden, cut out, and filled.

Takes in ←

- **the SAM 2 model** from [102]; **the frames** from [180]; **the person box** from [273].

Sends out →

- **the raw person stencil** to the grow/soften node [182].

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
individual_objects / settings (off)	Whether to track each object separately. Off = treat the detected person region as one shape.	For a single subject, off is correct and clean.	Off is slightly lighter.

[182] Grow + Soften the Stencil (GrowMaskWithBlur)

In plain words: Expands the person stencil outward a little and blurs its edge. Growing it guarantees the whole body is covered (no stray hair or fingertips peeking out); softening the edge means later blends look natural instead of showing a hard outline. This grown stencil (saved as ‘mask’) is reused all over the pipeline.

Takes in ←

- the raw stencil from SAM 2 [104].

Sends out →

- the grown, softened stencil, saved as ‘mask’ — used by the hide step, the cut-out step, and the background-fill step.

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
expand (4)	How many pixels to grow (+) or shrink (–) the stencil. Positive grows it outward.	Grow more = safer full coverage of the body, but eats into background. Too much = halo around the person.	Negligible.
blur_radius (0)	How much to soften the stencil edge.	More = smoother, more natural blends at the boundary. 0 = hard edge.	Negligible.
tapered_corners / fill_holes	Tidy-up options for the stencil shape (round corners, close gaps).	Cleaner, gap-free stencil.	Negligible.

[271] Solid Grey Image (EmptyImage)

In plain words: Creates a plain mid-grey image, the same length as the video. This is the colour the person will be painted over with. (Grey 128/128/128 — a neutral block.)

Takes in ←

- the frame count from the video loader, so it makes one grey frame per video frame.

Sends out →

- the grey frames to the hide step [191].

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
color (8421504 = grey)	The fill colour painted over the person. 8421504 is medium grey.	Changes how the hidden silhouette looks. Grey is neutral and non-distracting.	None.
width / height (640)	Size of the grey image (kept equal to the video).	Must match the video.	None.

[191] Hide the Person — “blur whole body” (ImageCompositeMasked)

In plain words: THE key privacy action of Tier 1. It paints the grey image over the original video, but ONLY inside the person stencil. Result: the real person becomes a flat grey silhouette while the background stays fully real. Note: despite the name ‘blur’, in this version it is a solid grey fill (a stronger form of hiding) because the separate Gaussian-blur node is switched off.

Takes in ←

- **destination** = the original frames (saved 'input_video').
- **source** = the grey frames from [271].
- **mask** = the person stencil from [182].

Sends out →

- the privacy video (real scene + grey person), saved as 'T1_blurred_video' — the foundation for Tiers 2A and 2B.

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
x / y (0, 0)	Offset of the grey layer. 0,0 = perfectly aligned.	Should stay 0 so the silhouette sits exactly on the person.	None.
resize_source (false)	Whether to rescale the grey layer. Off = it already matches.	Keep off.	None.

[173] Draw the Skeleton (DrawViTPose)

In plain words: Turns the raw skeleton key-points into a visible stick-figure image — coloured lines for limbs and dots for joints — one per frame. This stick figure is one of the two 'notes' sent to the cloud; it tells the generator exactly how the body should move, with zero identity information.

Takes in ←

- **skeleton data** from [273]; **frame size** from [180].

Sends out →

- the stick-figure pose frames, saved as 'pose_images' — sent to the cloud generator.

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
width / height (832×480)	Canvas size for the drawn skeleton.	Should match the working resolution so the pose lines up with the body.	Negligible.
point/line thickness (16)	How thick the drawn joints and bones are.	Thicker = clearer guidance for the generator; very thin can be missed.	None.

[192] Make the Face-Expression Mesh (SITARAFaceCanonicalizer)

In plain words: Takes the cropped face and converts it into an abstract, identity-free pattern that captures expression and head pose (smiling, blinking, mouth shape, tilt) but throws away the actual identity. This is your project's privacy-critical component: it lets the cloud copy the acting without ever seeing the face. It is the second 'note' sent to the cloud.

Takes in ←

- **the face crop** from [273].

Sends out →

- the canonical expression face, saved as ‘facemesh’ — sent to the cloud generator.

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
output_size (512)	Resolution of the produced face-mesh image.	Higher = finer expression detail captured. Lower = coarser.	Slightly slower at higher sizes.
thresholds (0.7 / 0.45)	Confidence needed to detect the face / accept its landmarks.	Higher = only confident, stable meshes (fewer false detections). Lower = more lenient but can produce noisy meshes on hard frames.	Negligible.

[201] (+[289]) Tier 1 Output (VHS_VideoCombine)

In plain words: Stitches the Tier 1 frames back into a downloadable video file so you can inspect what leaves this stage (the grey-silhouette result and review collages).

Takes in ←

- the assembled Tier 1 frames / collage.

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
frame_rate (10)	Playback speed of the saved file.	Should match the pipeline's fps.	None.
crf (19)	Video compression quality (0 = lossless, 51 = worst).	Lower = higher quality, bigger file. 19 ≈ visually lossless.	Lower = slightly slower to encode, larger files.
format / pix_fmt (h264-mp4 / yuv420p)	Codec and pixel format for broad compatibility.	Standard, plays everywhere.	None.

Tier 2 Part A — Mobile: separate the grey person and package it

Still on the trusted phone. This short stage splits the privacy video into ‘just the grey person’ versus ‘the scene’, and bundles the safe items for the cloud.

[206] Flip the Stencil (InvertMask)

In plain words: Takes the person stencil and flips it: white becomes black and black becomes white. So instead of selecting the person, it now selects everything except the person (the background).

Takes in ←

- **the person stencil** (saved ‘mask’).

Sends out →

- the inverted (background) stencil to [207].

No knobs.

[207] Keep Only the Grey Person (DrawMaskOnImage)

In plain words: Paints everything outside the person black, leaving just the grey person-shape floating on a black background. This isolated silhouette is what travels to the cloud as the body to animate.

Takes in ←

- **the privacy video** (saved ‘T1_blurred_video’); **the inverted stencil** from [206].

Sends out →

- the isolated grey person on black, saved as ‘C_blurredpart’ (and used as the base for the cloud’s hole/mask).

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
color (0,0,0 = black)	The colour painted into the masked-out region.	Black is the clean choice for isolating the subject.	None.

[212] Package to Cloud (VHS_VideoCombine)

In plain words: Saves the cloud package as a downloadable file: the grey person, plus (in the review collage) the pose and face-mesh — the three safe items the cloud is allowed to receive.

Settings are the same video-saving knobs as [201] (frame_rate, crf, format).

Tier 2 Part B-1 — Mobile: erase the person, fill the background

Still trusted. This branch produces a clean, person-free background by reconstructing what was hidden behind the subject. This is the branch the filename calls ‘b1 active.’

[263] Adjust the Stencil for Filling (GrowMaskWithBlur)

In plain words: Prepares the person stencil specifically for background filling — here it slightly shrinks the shape and softens it, so the fill blends cleanly with the true background right at the edges.

Takes in ←

- the person stencil (saved ‘mask’).

Sends out →

- the adjusted stencil to the background filler [215].

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
expand (–15)	Negative value shrinks the stencil inward by 15 px.	Shrinking avoids a faint outline of the person remaining at the very edge of the fill.	Negligible.
blur_radius (6)	Softens the stencil edge.	Smoother transition between filled and real background.	Negligible.

[215] Fill the Background — temporal median (SITARABackgroundFill)

In plain words: The clever background reconstructor and another of your project’s custom components. For each pixel hidden by the person, it looks across many frames of the video — frames where the person had moved away — and picks the most common (median) colour seen there. That reveals the true wall/floor/furniture behind the person. Where the person never moved (so the background was never seen), it leaves a small residual hole for the next node to patch.

Takes in ←

- the privacy video frames (saved ‘T1_blurred_video’); the adjusted stencil from [263].

Sends out →

- the filled background to the patch-filler [266].
- a residual mask (the bits it could not fill) to [267].

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
method (temporal_median)	How the hidden area is reconstructed. Temporal median = look across time and take the most common pixel. Excellent when the camera is fairly still and the person moves.	Temporal median gives clean, real backgrounds with no hallucination — as long as the person reveals the area at some point.	More frames analysed = slower; median over a clip is moderately heavy but runs on the phone tier.

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
temporal window (16)	How many neighbouring frames to look across when searching for the true background.	Bigger window = more chances to see behind the person = fewer leftover holes.	Bigger window = slower and more memory.
feather/blend (5)	How softly the filled region blends into the surrounding real pixels.	Higher = smoother seams.	Negligible.

[267] **Grow the Leftover-Hole Stencil (GrowMaskWithBlur)**

***In plain words:** Takes the small residual holes the median couldn't fill and grows + softens them slightly, so the patch-filler has a clean region to work on.*

Takes in ←

- **the residual mask** from [215].

Sends out →

- the prepared hole-stencil to the patch-filler [266].

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
expand (4) / blur (4)	Grow the leftover holes by 4 px and soften by 4 px.	Ensures the patch-filler covers the whole gap and blends.	Negligible.

[266] (+[264]) **Patch the Remaining Holes — LaMa (INPAINT_InpaintWithModel)**

***In plain words:** Uses the 'big-lama' AI fill model to intelligently paint in any background that was never revealed (e.g. an area the person stood in the entire time). It invents plausible background texture so the scene looks complete. The combination of 'copy real background where possible (median) + invent only where necessary (LaMa)' gives a clean result while keeping invention to a minimum.*

Takes in ←

- **the LaMa model** from [264]; **the filled background** from [215]; **the hole-stencil** from [267].

Sends out →

- the final clean background, saved as 'filled_background' — used at the very end to place the new person on.

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
seed (randomize)	Controls the random pattern LaMa invents for unseen areas.	Different seeds = slightly different invented texture. Fix it for repeatable results.	None.

[220] **Filled-Background Output (VHS_VideoCombine)**

In plain words: Saves the clean, person-free background as a downloadable file for inspection.

Same video-saving knobs as [201].

Tier 2 Part B-2 — Cloud (untrusted): generate the synthetic person

This is the powerful, GPU-heavy stage that runs in the cloud. It receives only the three safe items (grey person silhouette, pose, face-expression mesh) plus the reference photo, and from them generates a brand-new human performing the same motion and expression. It never sees the real face. This is also where almost all of the run-time is spent, so its knobs are the most important for the quality↔speed balance.

First, three small steps prepare the ‘where the body goes’ guide

[223] Silhouette → Stencil (ImageToMask)

In plain words: Converts the grey-person-on-black image back into a clean white-on-black stencil, marking exactly where the body should appear in the generated frame.

Takes in ←

- the isolated grey person (saved ‘C_blurredpart’).

Sends out →

- the body stencil to the grow node [224].

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
channel (red)	Which colour channel becomes the stencil.	Picks up the body region cleanly.	None.

[224] Grow the Body Stencil (GrowMaskWithBlur)

In plain words: Expands and softens the body stencil so the generated person has a little room and blends at the edges.

Takes in ←

- the body stencil from [223].

Sends out →

- the grown stencil to the pixelation node [225].

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
expand (6) / blur (4)	Grow by 6 px, soften by 4 px.	More room and smoother edges for the new body.	Negligible.

[225] Pixelate the Stencil — the active ‘blockify’ (BlockifyMask)

In plain words: Chops the body stencil into coarse square blocks (like big pixels). This deliberately throws away the fine outline of the original body so its exact shape cannot leak through; the generator only learns the rough region to fill. This is the pixelation step that is **STILL ON** — separate from the Tier 1 one named in the filename, which is off.

Takes in ←

- the grown stencil from [224].

Sends out →

- the blocky stencil to the cloud generator [62] (as its mask) and to the ‘black hole’ maker [242].

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
block_size (16)	Size of each square block in pixels.	Bigger blocks = coarser, more abstract body region (more privacy, less precise placement). Smaller = finer outline (more precise, less abstraction).	Negligible.

[242] **Punch a Black Hole (DrawMaskOnImage)**

In plain words: Paints the body region pure black on the silhouette image, creating an empty black hole exactly where the person should be. Feeding this ‘hole’ to the generator forces it to invent a FULL new person there from scratch (rather than just lightly editing the grey shape). This is what makes the output a genuinely new human.

Takes in ←

- **the isolated grey person** (saved ‘C_blurredpart’); **the blocky stencil** from [225].

Sends out →

- the black-hole background image to the cloud generator [62].

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
color (0,0,0 = black)	The fill colour of the hole.	Black forces full character generation. This is the intended privacy behaviour.	None.

Then the generator assembles everything and creates the video

[62] **Pack the Conditioning — the recipe (WanVideoAnimateEmbeds)**

In plain words: Gathers every ingredient the generator needs and packs them into a single ‘recipe’: the reference photo (who to look like), the pose (how to move), the face-mesh (how to emote), the black-hole background (where to put the new body), the stencil, and the size/length. Think of this as loading the brief before the AI starts painting.

Takes in ←

- **reference image** (saved ‘reference_image’), **pose** (‘pose_images’), **face-mesh** (‘facemesh’),
- **black-hole background** from [242], **the blocky stencil** from [225], the VAE translator, width/height/frame-count.

Sends out →

- the packed recipe to the sampler [27].

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
width / height (overridden → 640)	The canvas the new person is generated on (set by the master dials).	Bigger = sharper, more detailed person. Smaller = softer.	Cost grows ~quadratically; the dominant memory/time driver.
frames / window settings	How many frames and how they are batched.	More frames = longer clip; consistency handled by the context options below.	More frames = proportionally slower.
color-match flag	Whether to match the new person's colours to the reference.	On = more consistent tone with the reference photo.	Negligible.

[110] Long-Video Windowing (WanVideoContextOptions)

In plain words: Generators can only hold a limited number of frames in mind at once. This node tells the sampler to process a long clip in overlapping windows and stitch them, so motion stays smooth across the whole video instead of jumping at chunk boundaries.

Sends out →

- the windowing plan to the sampler [27].

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
context_frames (81)	How many frames are generated together in one window.	Bigger window = better temporal consistency (less flicker between frames).	Bigger window = more GPU memory and slower per window.
context_overlap (32)	How many frames neighbouring windows share.	More overlap = smoother seams between windows.	More overlap = more recomputation = slower.
context_stride (4) / schedule (static_standard)	Step between windows and the windowing strategy.	Tunes the consistency-vs-cost trade-off.	Affects how many windows run.

[27] Generate the Video — the sampler (WanVideoSampler)

In plain words: The actual AI 'painting' engine. Starting from random noise, it repeatedly refines the image over a number of steps until the new person emerges, guided by the recipe from [62] and the text guidance. This node, together with the resolution, decides almost the entire run-time and the look of the result.

Takes in ←

- the model** (with LoRAs and block-swap applied), **the packed recipe** from [62], **the text guidance** from [65].

Sends out →

- the generated frames in compressed form ('latent') to the decoder [28].

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
steps (6)	How many refinement passes. THE main quality↔speed dial for generation.	More steps = more detail and coherence, up to a point. 6 is low — only possible because the speed LoRA is loaded (see [171]).	Time scales almost linearly: 12 steps ≈ twice as slow as 6. Raise for final renders, keep low for tests.
cfg (1)	How strongly the text prompt is enforced.	With the distilled speed model, ~1 is correct. Higher can over-saturate or distort.	Higher cfg roughly doubles compute per step.
shift (5)	Shifts the noise schedule toward detail or motion.	Tunes the detail/motion balance; small changes only.	Negligible.
scheduler (dpm++_sde)	The maths recipe for the refinement steps.	Different schedulers trade off look, smoothness and how few steps are needed.	Some schedulers need more steps for the same quality.
seed (fixed)	The starting random pattern.	Same seed = identical result every run (great for comparing one changed knob). Randomize to explore different takes.	None.
force_offload (true)	Frees GPU memory between steps.	No quality change; prevents out-of-memory crashes on tight VRAM.	Slightly slower, but safer.

[28] Decode to Real Pixels (WanVideoDecode)

In plain words: The generator works in a compressed ‘sketch’ space to save memory. This node uses the VAE translator to turn that sketch back into a normal, full-colour video of the new person on a blank background.

Takes in ←

- the VAE translator (saved ‘VAE’); the compressed frames from [27].

Sends out →

- the finished character video, saved as ‘character_video’ — used by Tier 2C.

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
enable_vae_tiling (false)	Decode the image in small tiles to save GPU memory.	On = lower memory but can show faint tile seams and is slower. Off = cleanest and fastest if memory allows.	Off is faster; turn ON only if you hit out-of-memory at high resolution.
tile sizes (272 / 144 ...)	Tile dimensions — only used when tiling is ON.	Bigger tiles = fewer seams.	Bigger tiles = more memory per tile.

[171] Speed + Relight LoRAs (WanVideoLoraSelectMulti)

In plain words: Loads two small ‘skill patches’ (LoRAs) onto the main model. The first relights the new person to match the scene; the second — the lightx2v distillation LoRA — is what lets the sampler produce good results in just ~6 steps instead of 20–30. This single patch is the reason the whole generation is fast.

Sends out →

- the patched model down the chain to the sampler.

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
relight LoRA strength (1.0)	How strongly to relight the generated person to the scene.	Higher = stronger relighting match; too high can look unnatural.	None.
speed (distill) LoRA strength (1.2)	How strongly the few-step speed behaviour is applied.	Enables the low step count. Too high can wash out detail; too low and 6 steps won't be enough.	Indirectly huge: it is what makes 6-step (fast) generation viable.

[51] Block-Swap — fit the model in memory (WanVideoBlockSwap)

In plain words: The generator is large. Block-swap parks some of its internal layers in ordinary system RAM and shuttles them to the GPU only when needed, so the model fits on a smaller graphics card. It is a pure memory-vs-speed trade with no effect on the final look.

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
blocks_to_swap (25)	How many of the model's layers to offload to system RAM.	No quality effect at all.	More swapped = less GPU memory needed but slower (more shuttling). Fewer swapped = faster but needs more GPU memory. Lower this if your card has spare memory.

[230] (+[290]) Character-Video Output (VHS_VideoCombine)

In plain words: Saves the generated person (on a blank background) as a downloadable file for inspection. This is where you would spot the 'black halo' or 'muted expression' issues if they occur.

Same video-saving knobs as [201]. (One copy of this is set to 1 fps for frame-by-frame inspection.)

Tier 2 Part C — Mobile: composite the new person onto the clean background

Back on the trusted phone. The cloud returned a new person on a blank background; this stage cuts that person out and lays them onto the reconstructed empty background to produce the finished video.

[259] Find the Generated Person (PoseAndFaceDetection)

In plain words: Detects where the newly generated person is in each frame, producing a box around them. Needed because the generated frame must be cut out precisely before compositing.

Takes in ←

- the character video (saved 'character_video'); detector model; width/height.

Sends out →

- the person box to the stencil maker [262].

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
width / height (640)	Resolution to detect at.	Match the working size for accurate boxes.	Higher = slower.

[262] Stencil the Generated Person (Sam2Segmentation)

In plain words: Makes a precise cut-out stencil of the generated person, so only the new body (not its blank background) gets pasted into the final scene.

Takes in ←

- SAM 2 model; the character video; the person box from [259].

Sends out →

- the character stencil to the feather node [235].

Same automatic behaviour as the Tier 1 SAM 2 node.

[235] Feather the Character Edges (GrowMaskWithBlur)

In plain words: Softens the edges of the character stencil so the pasted person blends into the background instead of showing a hard cut-out line.

Takes in ←

- the character stencil from [262].

Sends out →

- the feathered stencil to the final composite [236].

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
expand (2) / blur (6)	Grow slightly, soften the edge by 6 px.	More blur = softer, more natural blend into the scene. Too much = faint outline/halo.	Negligible.

[236] Final Composite (ImageCompositeMasked)

In plain words: THE final assembly. It pastes the generated person (using the feathered stencil) on top of the reconstructed empty background, exactly where the original person was. The output is the finished, privacy-protected video frame.

Takes in ←

- **destination = the clean filled background** (saved 'filled_background'), **source = the character video** (saved 'character_video'), **mask = the feathered stencil** from [235].

Sends out →

- the final frames to the final video saver [241] and the comparison strip [247].

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
x / y (0, 0)	Offset of the pasted person.	Keep 0 so the new person lands exactly where the old one was.	None.
resize_source (false)	Whether to rescale the character before pasting.	Off = both already the same size.	None.

[241] FINAL Output Video, with audio (VHS_VideoCombine)

In plain words: Saves the finished video and re-attaches the original audio extracted way back at the video loader — this is the deliverable result of the whole pipeline.

Takes in ←

- the final frames from [236]; the original audio (saved 'input_audio').

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
frame_rate (10)	Playback speed.	Match the pipeline fps.	None.
crf (19)	Final video quality/compression.	Lower = higher quality, larger file. For a final deliverable you might lower to ~15.	Lower = slightly slower encode.
trim_to_audio (false)	Whether to cut the video to the audio length.	Keeps video/audio in sync if enabled.	None.

Evaluation Only — the all-stages comparison strip

This region is purely for testing and demos. It is NOT part of the privacy product (it even uses the original video for reference).

[247] Side-by-Side Strip (ImageConcatMulti)

In plain words: Stitches the different stage outputs next to each other so you can compare them in one video: original, grey-silhouette, filled background, generated person, and final. The fastest way to judge whether each stage is doing its job.

Takes in ←

- review collages of each stage plus the final composite.

Sends out →

- the combined strip to the comparison saver [248].

Setting (current value)	What it does — in plain words	Effect on quality	Effect on speed / time
inputcount (4)	How many panels to stitch together.	More panels = more stages visible at once.	Negligible.
direction (right)	Lay panels left-to-right (or down).	Cosmetic layout choice.	None.
match_image_size (on)	Normalises panel sizes so they line up.	Keeps the strip tidy.	None.

[248] Comparison Output (VHS_VideoCombine)

In plain words: Saves the side-by-side comparison as a downloadable video.

Same video-saving knobs as [201].

Part 5 — Master Control Panel (quick tuning guide)

If you only remember a handful of knobs, remember these. They give you the most control over the two things you care about: how good the result looks, and how long it takes. They are listed roughly in order of impact.

To make a fast test run (sacrifice quality for speed)

- **Cap the frames** — [63] frame_load_cap to e.g. 16. Instantly shortens every stage.
- **Lower the resolution** — [150]/[151] to e.g. 384 or 448. Generation cost falls with the square of the size.
- **Drop the steps** — [27] steps to 4. Generation time falls almost linearly.
- **Fix the seed** — [27] seed = fixed, so you can change one knob at a time and compare fairly.

To make a high-quality final run (sacrifice speed for quality)

- **Raise the resolution** — [150]/[151] to 640 or higher (watch GPU memory).
- **Raise the steps** — [27] steps to 8–12 for more detail and stability.
- **Use all frames** — [63] frame_load_cap = 0; consider a higher force_rate for smoother motion.
- **Lower the output compression** — [241] crf to ~15 for a crisper saved file.

If you run out of GPU memory

- **Increase** [51] blocks_to_swap (slower, but fits a smaller card).
- **Turn on** [28] enable_vae_tiling (slower decode, less memory).
- **Lower the resolution** [150]/[151], or reduce [110] context_frames.

The single biggest impact table

Knob	Node	Turn UP for	Turn DOWN for
Resolution	[150]/[151]	Sharper, more detailed person (quadratically slower)	Speed and lower memory
Sampler steps	[27]	More detail/coherence (linearly slower)	Faster generation
Frame cap	[63]	Longer / full clip	Quick tests
Context frames	[110]	Less flicker between frames	Less memory, faster
Blocks swapped	[51]	Fits smaller GPU (slower)	Faster (needs more GPU memory)
Output CRF	[241]	Smaller file (lower quality)	Higher quality (bigger file)

Part 6 — What Is Switched Off (dormant nodes)

These nodes sit on the canvas but are bypassed (turned off) in this version. They are alternative or older ways of doing a step, kept as spare parts. Knowing they are off prevents confusion about what is actually running.

Dormant node	What it would do if switched on	Why it is off here
[190] ImageBlur — ‘strong Gaussian blur’	Blur the person with a Gaussian blur instead of a solid grey fill.	This version hides the person with a solid grey silhouette (stronger privacy) via [191]+[271], so the blur path is unused. (This is why ‘blur’ in the names is now really a grey fill.)
[108] BlockifyMask (Tier 1)	Pixelate the person stencil already in Tier 1.	This is the ‘no blockify’ in the filename — Tier 1 pixelation is intentionally off. (The cloud-side pixelation [225] is still on.)
[252] DWPreprocessor + [253] ImageBlend	An alternative pose estimator (DWPose) blended over the main one.	The pipeline uses the RTM/ViTPose path [273]/[173] instead; this alternative is off.
[172] PoseAndFaceDetection (Tier 1 copy)	A second, alternative pose/face detector for Tier 1.	Redundant with [273]; off.
[258] ColorToMask	Build a stencil from a specific colour.	An alternative masking route not used in this branch; off.
[35] WanVideoTorchCompileSettings	Compile the model for a speed boost.	Off here (compilation can be finicky to set up); the model runs uncompiled.

Part 7 — Plain-English Glossary

Every technical term used in this document, explained simply.

Term	Plain meaning
Node	A box that does one job in the workflow.
Wire / link	A connection carrying one node's output into another's input.
Input	Something fed into a node by a wire.
Setting / widget / parameter	A value you type into a node — the 'knobs you can turn'.
Set / Get box	A tidiness trick: 'Set' saves a result under a name; 'Get' hands it back elsewhere. An invisible wire.
PORT OUT / PORT IN	Labels marking the exact data allowed to cross from a trusted device to the cloud.
Mask / stencil	A black-and-white shape that selects an area (white = selected). Used to act only on the person or only on the background.
Grow / dilate a mask	Make the selected shape bigger (or smaller, if negative).
Feather / blur a mask edge	Soften a stencil's border so blends look natural.
Blockify / pixelate	Replace fine detail with coarse squares, on purpose, to remove identifying shape.
Silhouette	The filled outline of the person (here, a grey shape).
Composite	Paste one image layer onto another using a stencil.
Temporal median	Look across many video frames and take the most common pixel — used to see the true background behind a moving person.
Inpaint	AI 'fill-in' of a missing area with plausible content (here, LaMa fills unseen background).
Reference image	The single photo the generator uses to decide what the synthetic person looks like.
Pose / skeleton	A stick-figure of joints showing how the body is positioned each frame.
Face mesh (canonicalized)	An abstract pattern of facial expression and head pose with the identity removed.
Model	A large trained AI file that performs a task (detect, segment, generate).
VAE	A 'translator' that converts between the generator's compressed sketch and real pixels.
Latent	The compressed 'sketch' form the generator works in to save memory.
Sampler	The engine that refines noise into an image over a number of steps.
Steps	How many refinement passes the sampler makes — more = better and slower.
CFG	How strongly the text prompt is enforced during generation.
Seed	The starting random number — same seed gives the same result.

Term	Plain meaning
Scheduler	The maths recipe controlling how the refinement steps progress.
LoRA	A small 'skill patch' added to a model. Here one relights the person; one (distillation) enables fast few-step generation.
Block-swap	Parking parts of a big model in system RAM to fit a smaller GPU (slower but lighter).
VAE tiling	Decoding the image in tiles to save GPU memory (slower, can seam).
Context window	Generating a long video in overlapping chunks so motion stays smooth.
CRF	A video-compression quality number — lower means higher quality and bigger files.
fps (force_rate)	Frames per second — fewer frames means a faster run.

— End of guide —